

Dr. Wei Guo

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Academic Positions

Hong Kong University of Science and Technology Postdoctoral Fellow. Hosts: Prof. Khaled B. Letaief and Prof. Jun Zhang	Sept. 2023 – Present Hong Kong SAR, China
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Education

The Chinese University of Hong Kong, Shenzhen (CUHK-Shenzhen) Ph.D., Computer and Information Engineering. Advisor: Prof. Chuan Huang	Jul. 2023 Shenzhen, China
University of Electronic Science and Technology of China (UESTC) B.Eng., Communications Engineering	Jun. 2017 Chengdu, China

Academic Visits

Purple Mountain Laboratories Visiting student.	Sept. 2021 – Nov. 2021 Nanjing, China
The Future Network of Intelligence Institute (FNii), CUHK-Shenzhen Visiting student. Advisors: Prof. Chuan Huang and Prof. Shuguang Cui	Jun. 2019 – Dec. 2019 Shenzhen, China
Shenzhen Reserach Institute of Big Data (SRIBD) Visiting student. Advisor: Prof. Shuguang Cui	Feb. 2017 – Jun. 2017 Shenzhen, China

Current Research Interests

Wireless Communications, Generative AI, Integrated Communications and AI

Publications (✉ denotes correspondence)

Journals:

- Y. Wang, **W. Guo**[✉], C. Huang[✉], G. Chen, Y. Zhang, and S. Cui, “[Point Cloud Environment-Based Channel Knowledge Map Construction](#),” *IEEE Trans. Cogn. Commun. Netw.*, early access, 2026.
- J. Tong, **W. Guo**, J. Shao, Q. Wu, Z. Li, Z. Lin, and J. Zhang, “[WirelessAgent: Large Language Model Agents for Intelligent Wireless Networks](#),” *China Commun.*, vol. 23, no. 3, pp. 265-285, Mar. 2026.
- T. Liao, **W. Guo**[✉], H. He, S. Song, J. Zhang, and K. B. Letaief, “[Joint Beamforming and Antenna Position Optimization for Fluid Antenna Assisted MU-MIMO Networks](#),” *IEEE J. Sel. Areas Commun. (JSAC)*, vol. 44, pp. 1209-1226, 2026.
- J. Shao, J. Tong, Q. Wu, **W. Guo**, Z. Li, Z. Lin, and J. Zhang, “[WirelessLLM: Empowering Large Language Models Towards Wireless Intelligence](#),” in *J. Commun. Inf. Netw. (JCIN)*, vol. 9, no. 2, pp. 99-112, Jun. 2024.
- W. Guo**, C. Huang, X. Qin, L. Yang, and W. Zhang, “[Dynamic Clustering and Power Control for Two-Tier Wireless Federated Learning](#),” in *IEEE Trans. Wireless Commun. (TWC)*, vol. 23, no. 2, pp. 1356-1371, Feb. 2024.

Updated on: October 9, 2026

- W. Guo, R. Li, C. Huang, X. Qin, K. Shen, and W. Zhang, “Joint Device Selection and Power Control for Wireless Federated Learning,” in *IEEE J. Sel. Areas Commun. (JSAC)*, vol. 40, no. 8, pp. 2395-2410, Aug. 2022.
- W. Guo, H. Zhang, and C. Huang, “Energy Efficiency of Two-Way Communications under Various Duplex Modes,” in *IEEE Internet Things J. (IoT-J)*, vol. 8, no. 3, pp. 1921-1933, Feb. 2021.

Magazines:

- B. Huang, Y. Wang, W. Guo[✉], Z. Guo, D. Wu, R. Li, D. Liu, W. Lan, C. Huang[✉], and S. Cui, “Building Low-Altitude Communication Networks: A Digital Twin-Based Optimization Framework,” *IEEE Wireless Commun. Mag.*, early access, 2026.
- Y. Wang, C. Huang[✉], S. Zhang, G. Chen, W. Guo[✉], S. Lan, L. Xu, X. Cheng, X. Tang, and S. Cui, “Towards Precise Channel Knowledge Map: Exploiting Environmental Information from 2D Visuals to 3D Point Clouds,” *IEEE Commun. Mag.*, early access, 2026.

Conference Proceedings:

- Y. Bian, W. Guo, Z. Wang, S. Song, J. Zhang, and K. B. Letaief, “GSBF: Gaussian Splatting for Environment-Aware Beamforming,” in *Proc. IEEE Glob. Commun. Conf. (Globecom)*, Dec. 2026.
- T. Liao, W. Guo, J. Qian, S. Song, J. Zhang, and K. B. Letaief, “Rethinking Mutual Coupling in Movable Antenna MIMO Systems,” in *Proc. IEEE Int. Conf. Commun. (ICC)*, May 2026.
- R. Li, Z. Xu, W. Guo, and Y.-J. Zhang, “Divergence Meets Dispersion: Efficient Wideband Beam Training for XL-MIMO,” in *Proc. IEEE Wireless Commun. Netw. Conf. (WCNC)*, Apr. 2026. **Best Paper Award**
- Y. Bian, W. Guo, J. Yang, S. Song, J. Zhang, S. Jin, and K. B. Letaief, “Multi-modal Data Driven Virtual Base Station Construction for Massive MIMO Beam Alignment,” in *Proc. IEEE Wireless Commun. Netw. Conf. (WCNC)*, Apr. 2026.
- E. Tang, W. Guo, H. He, S. Song, J. Zhang, and K. B. Letaief, “Accurate and Fast Channel Estimation for Fluid Antenna Systems with Diffusion Models,” in *Proc. IEEE Glob. Commun. Conf. (Globecom)*, Dec. 2025.
- T. Liao, W. Guo, H. He, S. Song, J. Zhang, and K. B. Letaief, “Fluid Antenna-Assisted MU-MIMO Systems with Decentralized Baseband Processing,” in *Proc. IEEE Glob. Commun. Conf. (Globecom)*, Dec. 2025.
- W. Guo, M. He, C. Huang, H. He, S. Song, J. Zhang, and K. B. Letaief, “Model-driven Deep Learning for Distributed Detection in WSNs with Binary Quantization,” in *Proc. IEEE Mediterranean Conf. Commun. Netw. (MeditCom)*, Jul. 2024.
- W. Guo, R. Li, C. Huang, X. Qin, K. Shen, and W. Zhang, “Optimized Device Selection and Power Control for Wireless Federated Learning,” in *Proc. IEEE Glob. Commun. Conf. (Globecom)*, Dec. 2022.
- W. Guo, C. Huang, X. Qin, L. Yang, and W. Zhang, “Dynamic Clustering and Power Control for Two-Tier Over-the-Air Federated Learning,” in *Proc. IEEE/CIC Int. Conf. Commun. China (ICCC)*, Aug. 2022.
- W. Guo and C. Huang, “Energy Efficiency of Full-Duplex Two-Way Channels,” in *Proc. Glob. Conf. Signal Inf. Process. (GlobalSIP)*, Nov. 2019.

Patents:

- CN patent: “一种面向无线联邦学习的用户协同传输方法”, 黄川, 崔曙光, 郭玮. 专利申请号: 202211032064.0, 2022.

Preprints:

- W. Guo, Y. Zhang, X. Ge, and J. Zhang, “Equal Path Cost, Unequal Output Effects: Understanding Perturbation Propagation in Diffusion Models,” [arXiv: 2610.11380](#), 2026.
- S. Xie, W. Guo[✉], S. Song, J. Zhang, Y.-J. Zhang, K. B. Letaief, “Task-Oriented Communication with Hybrid-Precision Models,” [arXiv: 2607.16766](#), 2026, submitted to *IEEE Trans. Mobile Comput.*
- Y. Bian, W. Guo, J. Yang, S. Song, J. Zhang, S. Jin, and K. B. Letaief, “Multi-Modal Environment-Aware Beam Management for Massive MIMO: A Geometry-Driven Virtual Base Station Framework,” [arXiv: 2606.26567](#), 2026, currently under **major revision**.
- E. Tang, W. Guo[✉], H. He, S. Song, J. Zhang, K. B. Letaief, “A Unified Two-Stage Generative Diffusion Framework for Channel Estimation and Port Selection in Multiuser MIMO-FAS,” [arXiv: 2605.29679](#), 2026, submitted to *IEEE Trans. Wireless Commun.*, currently under **major revision**.
- T. Liao, W. Guo[✉], J. Qian, Z. Wang, S. Song, J. Zhang, and K. B. Letaief, “Rethinking Mutual Coupling in Movable Antenna MIMO Systems: Modeling and Optimization,” [arXiv: 2604.26282](#), 2026, submitted to *IEEE Trans. Wireless Commun.*
- J. Tong, Z. Li, F. Liu, W. Guo, and J. Zhang, “WirelessAgent++: Automated Agentic Workflow Design and Benchmarking for Wireless Networks,” [arXiv: 2603.00501](#), 2026, submitted to *IEEE Trans. Wireless Commun.*, currently under **major revision**.

Awards

IEEE WCNC Best Paper Award	2026
Exemplary Reviewer of IEEE Wireless Communications Letters	2020

Teachings

Teaching assistant at CUHK-Shenzhen:

EIE4006 Performances Evaluation of Communication Networks	Sum’ 2020-21
CIE6126 Performances Analysis of Computer and Communication Networks	Sum’ 2020-21
EIE3001 Signals and Systems	Spr’ 2021

Academic Services

Conference Reviewers:

- IEEE Global Communications Conference (Globecom): 2019-2025
- IEEE/CIC International Conference on Communications in China (ICCC): 2019-2025
- IEEE Wireless Communications and Networking Conference (WCNC), 2024.
- IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), 2024.

Journal Reviewers:

- IEEE Transactions on Wireless Communications

- IEEE Transactions on Communications
- IEEE Journal on Selected Areas in Information Theory
- IEEE Wireless Communications Letters
- IEEE Systems Journal
- IEEE Transactions on Machine Learning in Communications and Networking
- IEEE Internet of Things Magazine
- China Communications
- Journal of Communications and Information Networks
- ...

Presentations

2024 IEEE MeditCom, Madrid, Spain, Oral presentation
2022 IEEE Globecom, Rio de Janeiro, Brazil, virtual presentation
2022 IEEE/CIC ICC, Foshan, China, Oral presentation
2019 IEEE GlobalSIP, Ottawa, Canada, Oral presentation